HUNADI SYLVIAH MOTUBATSE
BrightLearn Tutorials - Data Analytics
SQL Fundamentals
Exercise 3: SQL CASE STATMENT

Question 1

Syntax:

```
SELECT Product_Name,
       Price,
CASE
    WHEN price > 1000 THEN 'Expensive'
    WHEN Price BETWEEN 100 AND 1000 THEN 'Mid-range'
    WHEN Price < 100 THEN 'Budget'
    END AS Price_classification
    FROM Products;
```

OUTPUT TABLE

Product_Name	Price	Price_classification
Laptop	1200.00	Expensive
phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

Question 2

Syntax:

```
SELECT Customer_Name,
       amount,
CASE
    WHEN amount ≥ 1000 THEN 'High Value'
    WHEN amount BETWEEN 500 AND 99999 THEN 'Medium Value'
    WHEN amount < 500 THEN 'Low Value'
```

END AS Order_label
From orders ;

Output table

Customer_name	amount	Order_Category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

Question 3

Syntax:

```
SELECT      Customer-name, department
            Amount,
CASE
    WHEN department = 'IT' AND Salary > 80000 THEN 'SENIOR IT'
    WHEN department = 'HR' AND salary > 55000 THEN
    'Experienced HR'
    ELSE 'Staff'
    END AS Position_Ranking
From employees ;
```

OUTPUT TABLE

Employee_name	department	salary	Position_Rank
John	IT	85000	Senior IT
Sara	HR	60000	Exprienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

Question 4

Syntax:

```
SELECT Student_name,
       Score,
CASE
    WHEN Score ≥ 90 THEN 'A'
    WHEN Score BETWEEN 80 AND 89 THEN 'B'
    WHEN Score BETWEEN 70 AND 79 THEN 'C'
    WHEN Score BETWEEN 60 AND 69 THEN 'D'
    ELSE 'F'
    END AS Student_grades
FROM students;
```

output table

Student_name	Score	Student_grades
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

Question 5

Syntax:

```
SELECT delivery_id,
       delivery_time_Minutes,
CASE
    WHEN delivery_time_Minutes ≤ 30 THEN 'Fast'
    WHEN delivery_time_Minutes BETWEEN 31 AND 60
    THEN 'On Time'
    ELSE 'Late'
    END AS Delivery_Performance
FROM deliveries;
```

Output Table

delivery_id	delivery_time_minutes	Delivery Performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	late
5	100	late

Question 6

Syntax:

```
SELECT issue_type,
       Priority,
CASE
    WHEN Priority = 3 THEN 'High'
    WHEN Priority = 2 THEN 'Medium'
    WHEN Priority = 1 THEN 'Low'
    END AS Priority_level
FROM tickets;
```

Output Table

Issue_type	Priority	Priority_level
Login Issue	1	Low
Server down	3	High
Slow System	2	Medium
Email error	2	Medium
Password reset	1	Low

Question 7

Syntax: NB! $\% = \frac{Actual}{total} \times 100$

```
SELECT Student_id,
    ((days_Present/total_days)*100) AS Attendance%,
CASE
    WHEN Attendance% >= 90 THEN 'Excellent'
    WHEN Attendance% BETWEEN 75 AND 89 THEN 'Good'
    ELSE 'Needs Improvement'
    END AS Attendence_classification
FROM attendance;
```

Output Table

Student_id	Attendence%	Attendance_classification
1	$(\frac{45}{50} \times 100) = 90$	Excellent
2	$(\frac{30}{50} \times 100) = 60$	Needs Improvement
3	$(\frac{48}{50} \times 100) = 96$	Excellent
4	$(\frac{25}{50} \times 100) = 50$	Needs Improvement
5	$(\frac{50}{50} \times 100) = 100$	Excellent

Question 8

Syntax:

```
SELECT Product_id,
       Stock_qty,
CASE
    WHEN Stock_qty = 0 THEN 'Out of Stock'
    WHEN Stock_qty BETWEEN 1 AND 4 THEN 'Low stock'
    ELSE 'In Stock'
    END AS Stock_level
FROM product_inventory;
```

Output table

Product-id	Stock-qty	Stock level
1	5	In Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Question 8

Syntax:

```
SELECT Subjects
        enrolled_students,
CASE
   WHEN enrolled_student >= 25 THEN 'Large'
   WHEN enrolled_student BETWEEN 10 AND 24 THEN 'Media
   ELSE 'Small'
  END AS Class_Size
FROM classes;
```

Output table

Subject	Enrolled_student	Class_size
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

Question 10

Syntax:

```
SELECT Payment_id,
       amount,
       Payment_Method,
CASE
    WHEN Payment_Method = 'cash' AND amount ≥ 200
        THEN 'Eligible for Discount'
    ELSE 'Not Eligible'
    END AS Discount_Eligible
FROM Payments;
```

Output table

Payment_id	amount	Method_Payment	Discount_Eligible
1	50.00	Card	Not Eligible
2	200.00	Cash	Eligible for Discount
3	150.00	Card	Not Eligible
4	75.00	Paypal	Not Eligible
5	300.00	Cash	Eligible for Discount